Data Structures and Algorithms Laboratory

Assignment# 3 (1/2 Week(s))

1. Using quick sort algorithm, write C programs to determine the number of comparisons and swap operations when the pivot is chosen as (i) the first element, (ii) random element, and (iii) median-of-three. Take the elements as the digits of your mobile number and stored in a file, which is to be given as the command line argument.
2. Write C programs for heap sort using the input 26, 5, 77, 1, 61, 11, 59, 15, 48, 19, which are stored in a file. Determine the number of swap operations performed for the same.
3. Write a C program for merge sort using the elements given in problem 2.
4. Given input {5271, 1234, 6754, 4199, 6979, 1999, 2352} in a file and hash function h(x) = x mod 10. Show the results using (i) separate chaining hash table, (ii) open addressing hash table using linear probing, (iii) open addressing hash table using quadratic probing and (iv) open addressing hash table with second hash function h2(x) = 7 – (x mod 7).